
◆ Natural Language Processing (NLP) – Q&A

Q1: What is NLP? How is it used in Data Science?

A: Natural Language Processing (NLP) is a field of AI that enables machines to understand, interpret, and generate human language. In data science, it is used for sentiment analysis, chatbot development, document classification, keyword extraction, and more.

Q2: What is Tokenization?

A: Tokenization is the process of splitting text into individual units (tokens), such as words or subwords. It's often the first step in NLP pipelines.

Q3: What is Lemmatization vs. Stemming?

A:

- **Stemming** cuts off word endings ("playing" → "play"), but may result in non-words.
 - **Lemmatization** returns valid root words using vocabulary and morphology analysis.
-

Q4: Explain TF-IDF. Why is it used?

A: TF-IDF (Term Frequency-Inverse Document Frequency) is used to evaluate how important a word is in a document relative to a corpus. It helps reduce the weight of commonly used words and highlight rare yet significant terms.

Q5: What are word embeddings?

A: Word embeddings are dense vector representations of words that capture semantic meaning. Examples include Word2Vec, GloVe, and FastText. They help models understand word relationships.

Q6: How does BERT work?

A: BERT (Bidirectional Encoder Representations from Transformers) uses transformers to capture context from both directions (left and right). It's pre-trained on a masked language modeling task and fine-tuned for downstream tasks like Q&A or classification.

Q7: What is Named Entity Recognition (NER)?

A: NER identifies and classifies entities in text into categories like names, organizations, locations, etc. It's useful in extracting structured info from unstructured text.

Q8: What are stop words and why do we remove them?

A: Stop words are common words like "the", "is", "and" that are often removed to reduce noise and dimensionality.

◆ **Deep Learning – Q&A**

Q9: What is Deep Learning?

A: Deep learning is a subset of ML based on neural networks with many layers. It excels at learning patterns in large datasets, especially in NLP and computer vision.

Q10: What is the difference between CNN, RNN, and Transformer?

- **CNNs:** Best for spatial data like images.
 - **RNNs:** Handle sequences (e.g., time series, text).
 - **Transformers:** Use attention mechanisms; state-of-the-art for NLP tasks.
-

Q11: What is overfitting and how can you prevent it?

A: Overfitting occurs when a model performs well on training data but poorly on new data. Prevention techniques:

- Dropout
- Regularization (L1/L2)

- Cross-validation
-

Q12: Explain the concept of an activation function.

A: Activation functions introduce non-linearity in neural networks. Common ones:

- ReLU: $\max(0, x)$
 - Sigmoid: Outputs between 0 and 1
 - Tanh: Outputs between -1 and 1
-

Q13: What is backpropagation?

A: Backpropagation is the algorithm used to compute the gradient of the loss function with respect to each weight by applying the chain rule, enabling neural networks to learn.

Q14: What optimizers do you know?

A: Popular optimizers:

- **SGD:** Basic optimizer with fixed learning rate.
 - **Adam:** Combines momentum and RMSProp; adaptive learning.
 - **RMSProp:** Works well for non-stationary objectives.
-

Q15: What is transfer learning in NLP?

A: Transfer learning uses a pre-trained model (like BERT, GPT) and fine-tunes it on a new task with less data. It boosts performance and reduces training time.

Q16: Describe a NLP project you've done.

A: Example: "I built a sentiment analysis model using BERT on Twitter data. I preprocessed data with NLTK, used HuggingFace Transformers, and fine-tuned BERT with a classification head for positive/negative sentiment."

Q17: How do you handle imbalanced classes in NLP classification?

A:

- Use class weights
 - Oversampling / undersampling
 - Use metrics like F1-score instead of accuracy
-

Q18: How do you evaluate NLP models?

A:

- For classification: Accuracy, Precision, Recall, F1-score
 - For generation: BLEU, ROUGE
-